

ZAMZAM JIROW ABDI

Part D - Publishing to GitHub and Branch Workflow

- * mkdir creates folders.
- * cd moves into a folder.
- * touch creates empty files.
- * echo ... > README.md puts text into a file using output redirection.
- * ls displays files.
- * cat displays file contents.
- * find . -print displays the project contents recursively.

src contains the actual program code, tests contains code used to test the program, and docs contains documentation. Keeping them separate makes a project easier to understand, maintain, and expand.

```
/tests
/docs
/docs/README.md
/src
/src/config.py
/src/utils.py
/src/main.py
iohamed@Mohameds-MacBook-Air python_lab %
```

Part B: Git Initialization and First Commit

```

mohamed@Mohameds-MacBook-Air python_lab % git init
Reinitialized existing Git repository in /Users/mohamed/python_lab/.git/
mohamed@Mohameds-MacBook-Air python_lab % printf "__pycache__/\n*.pyc\n.env\n" > .gitignore
mohamed@Mohameds-MacBook-Air python_lab % cat .gitignore
__pycache__/
*.pyc
.env
mohamed@Mohameds-MacBook-Air python_lab % git add .
mohamed@Mohameds-MacBook-Air python_lab % git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   .gitignore
        new file:   docs/README.md
        new file:   src/config.py
        new file:   src/main.py
        new file:   src/utils.py

mohamed@Mohameds-MacBook-Air python_lab % git commit -m"set up python lab project structure"
[master (root-commit) 55bfb5f] set up python lab project structure
Committer: Mohamed <mohamed@Mohameds-MacBook-Air.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

5 files changed, 4 insertions(+)
create mode 100644 .gitignore
create mode 100644 docs/README.md
create mode 100644 src/config.py
create mode 100644 src/main.py
create mode 100644 src/utils.py
mohamed@Mohameds-MacBook-Air python_lab % git log --oneline
55bfb5f (HEAD -> master) set up python lab project structure
mohamed@Mohameds-MacBook-Air python_lab % █

```

.gitignore is a file that tells Git which files and directories should not be tracked or included in commits. In this project, it prevents Python cache folders, compiled Python files, and .env files from being added to the repository. This helps keep the repository clean and prevents unnecessary or sensitive files from being uploaded.

The __pycache__/ and *.pyc patterns are important because Python automatically creates cache folders and compiled Python files when programs are run. These files are generated automatically and are not normally needed in the source-code repository. Ignoring them keeps the Git repository organized and prevents unnecessary files from being tracked.

The .env pattern is also important because .env files may contain sensitive information such as passwords, API keys, or other private configuration details.

Git commit history shows how a project has developed over time. Each commit records a set of changes along with information such as the commit identifier, author, date, and commit message. This allows developers to see what changes were made, when they were made, and how the project has progressed from one stage to another.

Part C - Writing and Committing Python Code

utils.py

```
from utils import square, is_even, celsius_to_fahrenheit

number = float(input("Enter a number: "))

print("Square:", square(number))

if is_even(number):
    print("The number is even.")
else:
    print("The number is odd.")

print("Fahrenheit equivalent:", celsius_to_fahrenheit(number))
```

main.py

```
from utils import square, is_even, celsius_to_fahrenheit
```

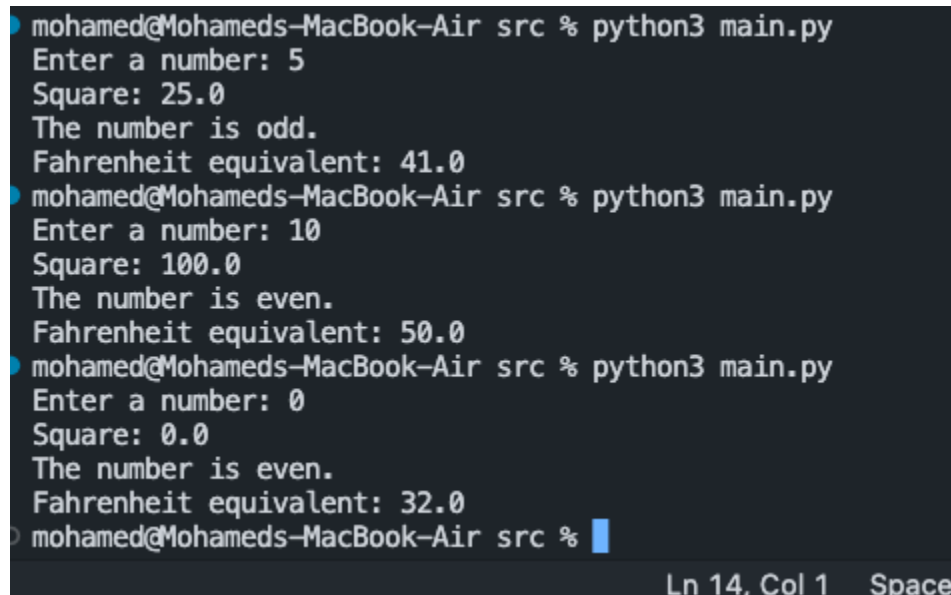
```
number = float(input("Enter a number: "))

print("Square:", square(number))

if is_even(number):
    print("The number is even.")
else:
    print("The number is odd.")

print("Fahrenheit equivalent:", celsius_to_fahrenheit(number))
```

Python's import system allows one Python file to use functions and other code defined in another Python file. In this project, the three functions `square`, `is_even`, and `celsius_to_fahrenheit` are defined in `utils.py`. The statement `from utils import square, is_even, celsius_to_fahrenheit` in `main.py` imports these functions so that they can be called from the main program. This keeps the project organized by separating reusable functions from the main program logic.



```
mohamed@Mohameds-MacBook-Air src % python3 main.py
Enter a number: 5
Square: 25.0
The number is odd.
Fahrenheit equivalent: 41.0
mohamed@Mohameds-MacBook-Air src % python3 main.py
Enter a number: 10
Square: 100.0
The number is even.
Fahrenheit equivalent: 50.0
mohamed@Mohameds-MacBook-Air src % python3 main.py
Enter a number: 0
Square: 0.0
The number is even.
Fahrenheit equivalent: 32.0
mohamed@Mohameds-MacBook-Air src %
```

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Part D - Publishing to GitHub and Branch Workflow

 zamzamjirow / python-lab

Q Type [Z] to search

[Code](#) [Issues](#) [Pull requests](#) 1 [Actions](#) [Projects](#) [Wiki](#) [Security and quality](#) [Insights](#)

 **python-lab** Public  P

 **Your main branch isn't protected** Dismiss Protect this branch
Protect this branch from force pushing or deletion, or require status checks before merging. [View documentation.](#)

 main  2 Branches  0 Tags T Add file <> Code

 **Mohamed and Mohamed** add python utility functions and main program 746f2e8 · 3 days ago ⌚ 2 Commits

 docs	set up python lab project structure	5 days
 src	add python utility functions and main program	3 days
 .gitignore	set up python lab project structure	5 days

 **README**

my python lab project

add personalized greeting feature #1



Open zamzamjirow wants to merge 1 commit into `main` from `feature/add-greeting`

Conversation 0 Commits 1 Checks 0 Files changed 2



zamzamjirow commented 1 minute ago

Owner ...

No description provided.



add personalized greeting feature

988de22



No conflicts with base branch
Merging can be performed automatically.

Merge pull request You can also merge this with the command line. [View command line instructions.](#)

Still in progress? [Convert to draft](#)



Add a comment

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one—[assign yourself](#)

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merged